

Final Project

Project Title

Design and Implementation of a Secure Azure Cloud Environment

Scenario

ABC Technologies is migrating its on-premises infrastructure to Microsoft Azure. The organization requires a secure cloud environment that protects virtual machines, storage accounts, and user identities while providing monitoring, threat detection, access control, and vulnerability assessment capabilities.

As a Cloud Security Engineer, your responsibility is to design, deploy, secure, monitor, and document a cloud environment following Azure security best practices.

The final environment should demonstrate preventive, detective, and monitoring security controls.

Project Objectives

Upon completion of this project, students should be able to:

- Deploy and secure Azure resources.
 - Implement access control using RBAC.
 - Secure Azure Storage using SAS and network restrictions.
 - Monitor resources using Azure Monitor.
 - Configure threat detection using Microsoft Sentinel.
 - Implement Just-In-Time access.
 - Perform basic vulnerability assessment using Nmap.
 - Produce security documentation and architecture diagrams.
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Task 1: Secure Virtual Machine Deployment

Deploy a Linux Virtual Machine and implement secure access controls.

Requirements

- **Create a Linux Virtual Machine.**
- **Configure SSH on custom port 2244 instead of the default port 22.**
- **Create and configure NSG rules.**
- **Allow only required inbound access.**
- **Verify SSH connectivity through port 2244.**

Evidence Required

- **VM deployment screenshot**
 - **NSG configuration screenshot**
 - **Successful SSH connection screenshot**
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Task 2: Identity and Access Management (RBAC)

Implement Role-Based Access Control using the Principle of Least Privilege.

Create Two Users

User 1 – VM Administrator

Responsibilities:

- **Manage Virtual Machine**
- **Start/Stop VM**
- **Modify VM settings**

Assigned Role:

- **Virtual Machine Contributor**
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User 2 – Storage Administrator

Responsibilities:

- **Manage Storage Account**
- **Upload and download blobs**
- **Generate SAS Tokens**

Assigned Role:

- **Storage Blob Data Contributor**
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Validation

Demonstrate that:

- **VM Administrator cannot manage storage resources.**
- **Storage Administrator cannot manage VM resources.**
- **Each user has only the permissions required for their responsibilities.**

Evidence Required

- **User creation screenshots**
 - **Role assignment screenshots**
 - **Access validation screenshots**
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Task 3: Secure Storage Configuration

Implement secure Azure Blob Storage.

Requirements

- **Create a Storage Account.**
- **Create a Private Blob Container.**
- **Upload sample files.**
- **Disable anonymous/public access.**
- **Generate SAS Token for temporary access.**
- **Configure IP-based network restrictions.**

Network Restriction

Allow storage access only from a specific public IP address.

Validation

Demonstrate:

- **Public access blocked.**
- **SAS URL access working.**
- **Unauthorized IP access denied.**

Evidence Required

- **Storage configuration screenshots**
 - **SAS token demonstration**
 - **IP restriction configuration screenshots**
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Task 4: Monitoring and Dashboard Visualization

Implement centralized monitoring.

Requirements

Configure:

- **Azure Monitor**
- **Log Analytics Workspace**
- **VM Insights**

Generate:

- **CPU Activity**
- **Memory Activity**
- **Network Activity**
- **Disk Activity**

Dashboard Requirements

Create a dashboard displaying:

- **CPU Usage**
- **Memory Usage**
- **Network Traffic**
- **Disk Activity**

Use visualizations such as:

- **Line Charts**
- **Gauge Panels**
- **Single Value Statistics**

Validation

Generate CPU load on the VM and demonstrate real-time metric updates.

Evidence Required

- **Azure Monitor screenshots**
 - **Dashboard screenshots**
 - **CPU utilization demonstration**
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Task 5: Microsoft Sentinel Integration

Implement Security Information and Event Management (SIEM).

Requirements

Configure:

- Log Analytics Workspace
- Microsoft Sentinel
- Syslog/Security Event Collection

Threat Detection Scenario

Generate multiple failed SSH login attempts against the VM.

Alert Rule

Create an alert that triggers when:

- More than five failed SSH login attempts occur.

Validation

Demonstrate:

- Failed login logs
- Sentinel query results
- Generated alert or incident

Evidence Required

- Sentinel configuration screenshots
 - KQL query screenshots
 - Alert screenshots
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Task 6: Just-In-Time (JIT) Access

Implement temporary access control using Microsoft Defender for Cloud.

Requirements

- Enable JIT Access.
- Configure Port 80 for JIT protection.
- Request temporary access.
- Verify automatic closure after expiration.

Validation

Demonstrate:

- Access request

- Temporary access approval
- Automatic removal of access

Evidence Required

- JIT configuration screenshots
 - Access request screenshots
 - Access expiration screenshots
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Task 7: Vulnerability Assessment

Perform a basic vulnerability assessment using Nmap.

Requirements

Use Nmap to:

- Identify open ports
- Detect running services
- Analyze exposed attack surface

Validation

Document:

- Open ports discovered
- Services identified
- Security observations

Evidence Required

- Nmap scan results
 - Service enumeration screenshots
 - Findings summary
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Task 8: Security Assessment Report

Prepare a final report containing:

Risks Identified

Examples:

- Open services
- Weak configurations
- Access control issues
- Public exposure risks

Security Controls Implemented

Include:

- RBAC
- NSG Rules
- Custom SSH Port
- Private Blob Storage
- SAS Token Access
- IP Restrictions
- Azure Monitor
- Microsoft Sentinel
- JIT Access

Recommendations

Provide recommendations to further improve the security posture of the environment.

Architecture Diagram

Create an architecture diagram showing:

- Resource Group
 - Virtual Machine
 - NSG
 - Azure Monitor
 - Log Analytics Workspace
 - Microsoft Sentinel
 - Storage Account
 - Private Blob Container
 - SAS Access
 - RBAC Users
 - JIT Access
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Final Deliverables

Submit the following:

- **Architecture Diagram**
 - **VM Configuration Screenshots**
 - **RBAC Configuration Screenshots**
 - **Storage Security Screenshots**
 - **Azure Monitor Dashboard Screenshots**
 - **Microsoft Sentinel Screenshots**
 - **JIT Access Screenshots**
 - **Nmap Scan Results**
 - **Security Assessment Report (5–10 Pages)**
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Submission Notes

- **All screenshots must be properly labelled.**
- **Security controls should be clearly explained.**
- **Students should include observations and findings.**
- **Evidence should demonstrate successful implementation of each task.**